

Analysis of the Obstacle Potential Function

1 Potential Function

$$U_{obst} = \frac{\lambda(|\Delta v|/\Delta v_{max})}{k(\Delta x, \Delta y, \Delta v_x, \Delta v_y)} \quad (1)$$

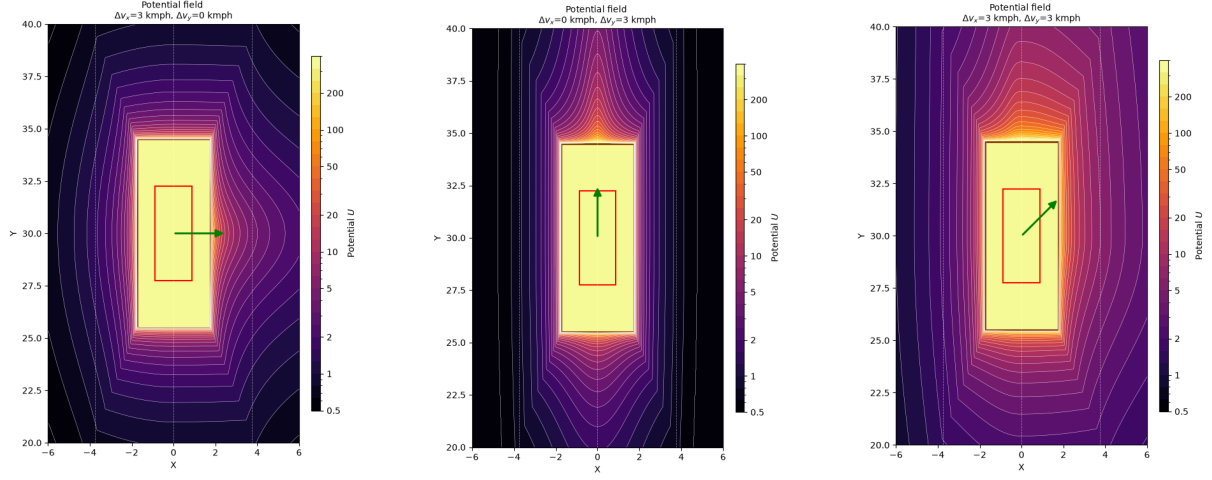
$$k(\Delta x, \Delta y, \Delta v_x, \Delta v_y) = \begin{cases} 0, & \text{if } |\Delta y| < l \text{ and } |\Delta x| < w \\ \left(1 - \frac{w}{|\Delta x|}\right) \sqrt{\left(\frac{\Delta x}{\tau_x}\right)^2 + \left(\frac{\Delta y}{\tau_y}\right)^2}, & \text{if } |\Delta y| \leq \frac{l|\Delta x|}{w} \\ \left(1 - \frac{l}{|\Delta y|}\right) \sqrt{\left(\frac{\Delta x}{\tau_x}\right)^2 + \left(\frac{\Delta y}{\tau_y}\right)^2}, & \text{otherwise} \end{cases} \quad (2)$$

$$\tau(\Delta x, \Delta v_x) = \frac{(1 + \alpha|\Delta v_x|) + (\alpha|\Delta v_x| - 1) \tanh(-\Delta x \Delta v_x)}{2} \tau_x \quad (3)$$

$$\tau(\Delta y, \Delta v_y) = \frac{(1 + \alpha|\Delta v_y|) + (\alpha|\Delta v_y| - 1) \tanh(-\Delta y \Delta v_y)}{2} \tau_y \quad (4)$$

Here $\Delta x, \Delta y$ are the relative position wrt obstacle vehicle, $\Delta v_x, \Delta v_y$ the relative velocities wrt obstacle vehicle.

1.1 Potential Field Plots



(a) Varying Δv_x

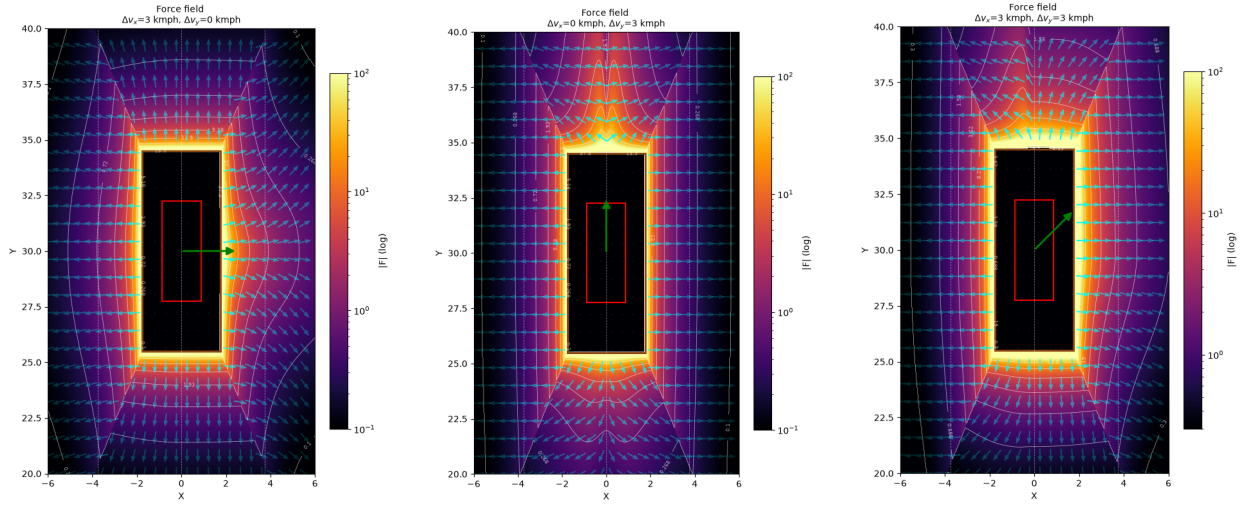
(b) Varying Δv_y

(c) Varying $\Delta v_x, \Delta v_y$

Figure 1: Obstacle potential field U_{obst} .

1.2 Force Field

$$F = -\nabla U_{obst}$$



(a) Varying Δv_x

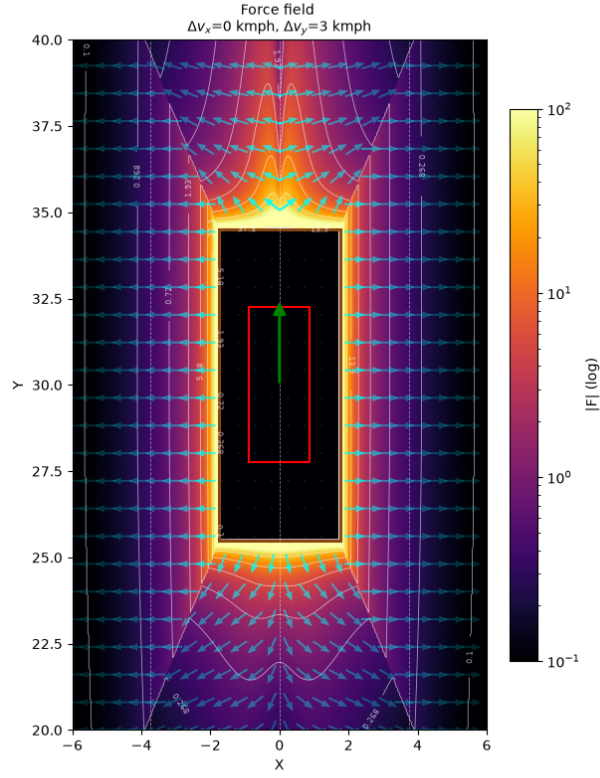
(b) Varying Δv_y

(c) Varying $\Delta v_x, \Delta v_y$

Figure 2: Force field derived from U_{obst} .

2 Possible Issues

2.1 Anomalous dip in the magnitude of force field



The magnitude of the force field dips near $x=0$ in the above plot which is undesirable as the magnitude of force field should be maximum along $x=0$ when compared to the neighbouring points.

2.2 Sensitivity to τ_x, τ_y and Parallel Forces

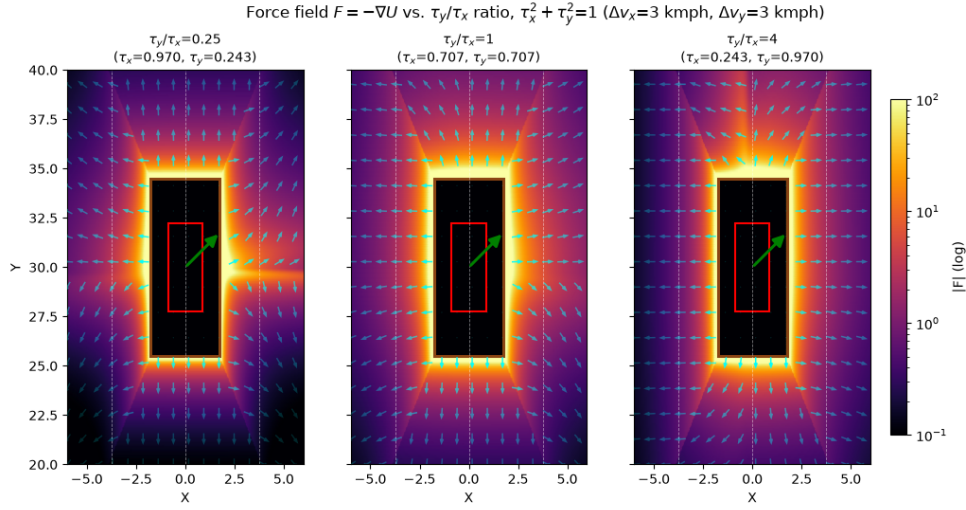


Figure 3: Sensitivity of the field to τ_x, τ_y .

The shape of the potential is highly sensitive to τ_y/τ_x and very high value or a very small value can cause the force field to be parallel in certain region which is not desirable.

2.3 Velocity Behavior at a fixed point

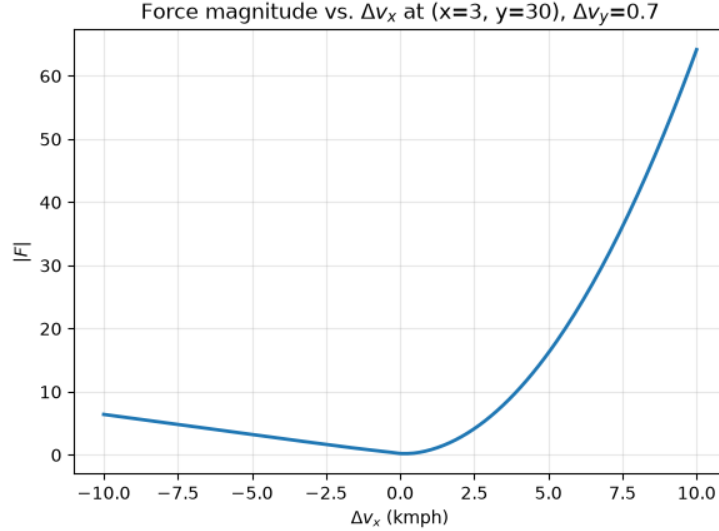


Figure 4: Velocity at a fixed point.

The force increases with increase in receding velocity at a fixed point. In an ideal scenario the force should decrease when the vehicles are receding.

2.4 More free parameters than DOF

The model introduces 4 free parameters ($\lambda, \alpha, \tau_x, \tau_y$) while the degree of freedom of the model is only 3. So reduction of parameters will help in the optimization problem.